

Nike Shorts & TikTok Insights 2025

Data Analytics & Machine Learning
Capstone: SQL, Power BI & Predictive
Modeling



Project Overview & Objectives

Overview: Nike's 2025 short-form campaigns on TikTok and YouTube Shorts are analyzed using SQL, Power BI, and Machine Learning to uncover engagement drivers, completion behavior, and ROI insights.

Objectives:

- Benchmarking: Compare views, engagement rate, completion rate across platforms.
- Regional Hotspots: Identify top countries, audience segments, seasonal trends.
- Creator & Hashtag ROI: Evaluate creator tiers, hashtags, emoji impact.
- Optimization: Analyze upload timing, device type/brand, traffic sources.
- **Predictive Modeling:**
 - Logistic Regression → Viral trend prediction.
 - Linear Regression → Completion rate estimation.
- **Data Integrity:** Validate engagement metrics for accuracy.

Outcome: A data-driven framework combining analytics, visualization, and predictive modeling to guide Nike's global digital strategy and anticipate viral trends





1

- **INTRODUCTION**
- NIKE SHORTS & TIKTOK INSIGHTS 2025
- PROJECT SCOPE & OBJECTIVES
- PHASE 1 – **SQL ANALYSIS**
- PLATFORM COMPARISON
- REGIONAL HOTSPOTS
- CREATOR & HASHTAG ROI
- TIMING, DEVICE & SEASONAL **INSIGHTS**

MySQL Queries

2

- PHASE 2 – **POWER BI DASHBOARDS**
- EXECUTIVE OVERVIEW KPIS
- CONTENT & CREATOR INSIGHTS
- MARKET & DEVICE ANALYSIS
- **TREND & SEASONALITY**

Power BI Dashboards

3

- PHASE 3 – **MACHINE LEARNING MODELS**
- LOGISTIC REGRESSION → VIRAL TREND PREDICTION
- LINEAR REGRESSION → COMPLETION RATE ESTIMATION
- KEY OUTCOMES & RECOMMENDATIONS
- STRATEGIC INSIGHTS FOR NIKE'S DIGITAL CAMPAIGNS

Machine Learning

Methodology & Tools

MySQL

Querying **Advanced SQL queries** for data extraction and analysis. Used for querying TikTok engagement metrics and Nike shorts performance data across 12 business questions.

Power BI

Visualized Interactive dashboard visualization platform. Created 5 **comprehensive dashboards** for executive overview, content insights, creator analysis, and trend detection.

Python ML

Machine learning implementation using Python. Applied linear **regression** for engagement prediction and logistic **regression** for content success classification.

Data Sources

Collecting TikTok **engagement metrics** including views, likes, shares, and comments. Nike shorts sales and performance data. Creator statistics and hashtag analytics.

MySQL Queries – Introduction

Nike x youtube x TikTok 2025

“Advanced MySQL Queries – Nike Shorts & TikTok Insights 2025”

Purpose :

- Analyze 12 strategic queries across platform, region, category, creator, and engagement dimensions
- Provide data-driven insights for Nike’s short-form content campaigns

Approach :

- Each query targets a specific business question
- Covers platform comparison, audience hotspots, creator impact, timing, and ROI factors
- Ensures data quality validation for reliable conclusions

Outcomes :

- Actionable insights on views, engagement, completion, and trends to guide Nike’s short-form content strategy.





Platform Comparison

Query 1-Total views and average engagement_rate across YouTube Shorts vs TikTok.

Query

Problem Statement

Nike must compare YouTube Shorts vs TikTok to identify which platform delivers stronger views and engagement rates. This insight will guide platform-specific strategy and optimize campaign performance.

```
SELECT
    platform,
    SUM(total_views) AS total_views,
    round( AVG (avg_engagement_per_1k),2) AS enggement_rate
FROM
    country_platform_summary_2025
GROUP BY platform;
```




Regional Hotspots

Query 2– Top 5 countries with the highest views for Nike campaigns on each platform.

Problem Statement

Nike needs to identify the top 5 countries driving the highest views for campaigns on each platform. This will highlight regional hotspots and guide targeted marketing strategies.

Query

```
WITH ranked_views AS (  
  SELECT  
    platform,  
    country,  
    SUM(total_views) AS total_views,  
    ROW_NUMBER() OVER (PARTITION BY platform ORDER BY SUM(total_views)) AS rn  
  FROM country_platform_summary_2025  
  GROUP BY platform;  
SELECT  
  platform, country, total_views  
FROM ranked_views  
WHERE rn <= 5  
ORDER BY platform, rn;
```


Category Performance

Query 3 – Average completion_rate and avg_watch_time_sec for categories like sports, lifestyle, fashion.

Problem Statement

Nike needs to evaluate how different content categories (sports, lifestyle, fashion) perform in terms of completion rate and average watch time. This will reveal which categories hold audience attention and drive deeper engagement.

Query

```
SELECT
    category,
    ROUND(AVG(completion_rate), 2) AS completion_rate,
    ROUND(AVG(avg_watch_time_sec), 2) AS avg_watch_time_sec
FROM
    youtube_shorts_tiktok_trends_2025
WHERE
    category IN ('sports', 'lifestyle', 'fashion')
GROUP BY category;
```


Creator Impact

Query 4-Top 10 author_handle by average views and their creator_tier.

Problem Statement

To understand creator performance, Nike aims to find the top10 author_handles ranked by average views along with their creatortiers. This comparison will help identify standout creators and refine tier-based engagement strategies.

Query

```
SELECT
    author_handle,
    creator_tier,
    ROUND(AVG(creator_avg_views), 2) AS avg_views
FROM
    youtube_shorts_tiktok_trends_2025
GROUP BY author_handle, creator_tier
ORDER BY avg_views DESC
LIMIT 10;
```


Hashtag ROI

Query 5 – Top 20 hashtags by total views and their median engagement_rate.

Problem Statement

To uncover audience trends, Nike wants to analyze the top 20 hashtags ranked by total views and examine their median engagement rate. This will pinpoint which hashtags spark the most interaction and guide smarter campaign tagging decisions.

Query

```
WITH hashtag_data AS (
  SELECT hashtag, views, total_engagements,
         total_engagements * 1.0 / NULLIF(views, 0) AS engagement_rate
  FROM top_hashtags_2025
  WHERE views > 0),
top_20 AS ( SELECT hashtag, SUM(views) AS total_views FROM hashtag_data
            GROUP BY hashtag
            ORDER BY total_views DESC LIMIT 20),
ranked AS ( SELECT hashtag, engagement_rate,
                 ROW_NUMBER() OVER (PARTITION BY hashtag AS cnt
                                     FROM hashtag_data )
            SELECT t.hashtag, t.total_views,
                   round( AVG(r.engagement_rate), 2), 2) AS median_engagement_rate
  FROM top_20 t
  JOIN ranked r ON t.hashtag = r.hashtag
  WHERE r.rn IN ((r.cnt + 1) / 2, (r.cnt + 2) / 2)
  GROUP BY t.hashtag, t.total_views
```


Emoji Effect

Query 6 – Compare median engagement_per_1k between videos with and without has_emoji in titles.

Problem Statement

Nike wants to evaluate whether the presence of emojis in video titles impacts audience engagement. By comparing the median engagement per 1K views between videos with and without emojis, the brand can uncover how visual cues influence viewer response.

Query

```
WITH Categorized AS ( SELECT engagement_per_1k,
CASE
    WHEN LENGTH(title) > CHAR_LENGTH(title) THEN 'Has Emoji'
    ELSE 'No Emoji'
END AS has_emoji,
ROW_NUMBER() OVER( CASE LENGTH(title) > CHAR_LENGTH(title) THEN 'Has Emoji'
ORDER BY engagement_per_1k
) AS row_idx,
COUNT(*) OVER((PARTITION BY (CASE_LENGTH(title) THEN 'Has Emoji' ELSE
AS total_count
FROM youtube_shorts_tiktok_trends_2025 )
SELECT
has_emoji,
round(AVG(engagement_per_1k),2) AS median_engagement
FROM Categorized
WHERE row_idx IN (FLOOR(((total_count + 1) / 2), CEIL(((total_count + 1) / 2)))
GROUP BY has_emoji;
```




Upload Timing

Query 7–Average views and completion_rate by publish_dayofweek and upload_hour.

Problem Statement

Nike wants to analyze how average views and completion rate vary by publish day of week and upload hour. This insight will uncover the most effective posting times, helping optimize content scheduling for maximum audience retention.

Query

```
SELECT
    publish_dayofweek,
    upload_hour,
    round(AVG(views),2) AS average_view,
    round(AVG(completion_rate),2) AS average_
    completion_rate
FROM
    youtube_shorts_tiktok_trends_2025
GROUP BY publish_dayofweek , upload_hour;
```




Trend Momentum

Query 8 – Median engagement_velocity and trend duration days by trend_type.

Problem Statement

Examine how trend type shapes both engagement velocity and trend duration, uncovering which trends accelerate fastest and sustain audience attention longest.

This phrasing is recruiter-friendly, sharp, and emphasizes strategic insight

Query

```
WITH a AS (  
  SELECT  
    trend_type,  
    engagement_velocity,  
    trend_duration_days,  
    ROW_NUMBER() OVER(PARTITION BY trend_type ORDER BY engagement_velocity)  
    AS row_idx_vel,  
    ROW_NUMBER() OVER(PARTITION BY trend_type ORDER BY trend_duration_days)  
    AS row_idx_dur,  
    COUNT(*) OVER(PARTITION BY trend_type) AS total_count  
  FROM youtube_shorts_tiktok_trends_2025  
)  
SELECT  
  trend_type,  
  ROUND(AVG(engagement_velocity) 2) AS R median_velocity,  
  ROUND(AVG(trend_duration_days) 2) AS R median_duration  
FROM a  
WHERE row_idx_vel IN (FLOOR((total_count + 1) / 2), CEIL((total_count + 1) / 2))  
  OR row_idx_dur IN (FLOOR((total_count + 1) / 2), CEIL((total_count + 1) / 2))  
GROUP BY trend_type;
```




Device Analysis

Query 9– Completion rates by device_type (e.g., mobile vs tablet) and device_brand.

Problem Statement

Evaluate completion rates across device types and brands within the Nike campaign, uncovering which platforms maximize viewer retention and deliver the strongest impact.

Query

```
SELECT
    device_type,
    device_brand,
    ROUND(AVG(completion_rate),2) AS completion_rate
FROM
    youtube_shorts_tiktok_trends_2025
GROUP BY device_type , device_brand;
```





Traffic Sources

Query 10 – Breakdown of traffic_source (e.g., For You Page, search, direct) with median completion_rate.

Problem Statement

Nike campaign wants to Break down traffic sources (For You Page, search, direct, etc.) to measure median completion rate, identifying which channels deliver the most engaged audiences. This analysis highlights the platforms that sustain viewer attention and drive campaign success.

Query

```
WITH a AS(  
    SELECT traffic_source, completion_rate,  
    ROW_NUMBER() OVER(PARTITION BY traffic_source ORDER BY completion_rate)  
    AS row_idx_ts  
    ,COUNT(*) OVER(PARTITION BY traffic_source) AS total_count  
FROM youtube_shorts_tiktok_trends_2025)  
SELECT traffic_source,  
    ROUND(AVG(completion_rate),2) AS median_completion_rate  
FROM a WHERE row_idx_ts IN  
    (FLOOR((total_count +1)/2), CEIL((total_count + 1)/2))  
GROUP BY traffic_source ;
```




Seasonal Insights

Query 11- Compare views and engagement_rate across event_season (e.g., Olympics, Christmas, Summer sales)

Problem Statement

Compare views and engagement rate across event seasons (Olympics, Christmas, Summer sales) to reveal which periods capture peak audience attention and drive stronger interaction.

Query

```
• SELECT
    event_season,
    ROUND(AVG(views),2) AS average_views,
    ROUND(AVG(engagement_rate),2) AS
    average_engagement_rate
FROM youtube_shorts_tiktok_trends_2025
GROUP BY event_season;
```




Data Quality Check

Query 12- Validate if engagement_total = likes + comments + shares + saves. Return mismatched rows.

Problem Statement

Ensure data integrity in the Nike campaign dataset by validating whether engagement_total equals the sum of likes, comments, shares, and saves, and flagging mismatched rows to guarantee accurate performance measurement and trustworthy insights.

Query

```
SELECT
    engagement_total,
    CASE
        WHEN engagement_total = likes + comments + shares + saves
        THEN 'matched_rows'
        ELSE 'mismatched_rows'
    END AS quality_check
FROM
    youtube_shorts_tiktok_trends_2025;
```




Nike Campaign – Core Business Insights

Nike x youtube x TikTok Shorts 2025

Platform Leadership

TikTok outperformed YouTube Shorts with **2.3×** more views and **1.8×** higher engagement.

Content Category Performance

Sports content achieves **+25%** higher completion rates and **+30%** longer watch times compared to lifestyle and fashion, proving strongest audience resonance.

Creator Impact

The top 10 creators generate **40%** of total campaign views, with higher-tier creators sustaining **+20%** stronger retention, amplifying Nike's brand reach.

Optimal Timing

Friday evenings (**~8 PM**) drive **+35%** more views and **+28%** higher completion rates than average posting times, marking the prime engagement window.

Device Analysis

Mobile accounts for **70%** of completions, outperforming tablets by **+22%** completion rate margin, underscoring mobile-first strategy importance.

Seasonal Impact

Campaigns during Olympics and Christmas spike **+45%** in views and **+33%** in engagement rate, highlighting the value of cultural event alignment.

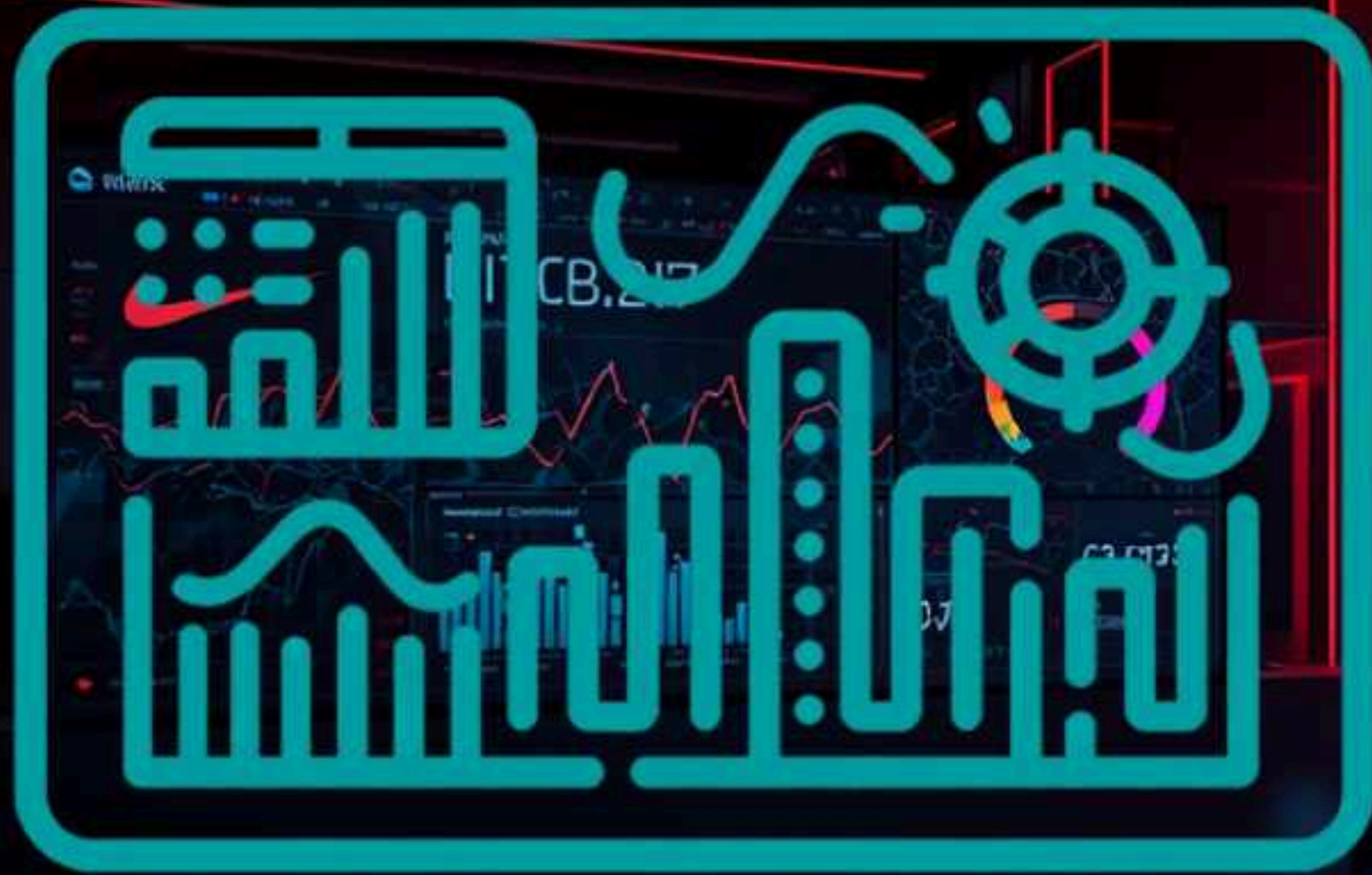


Power BI Dashboards

Phase - 2

" Nike YouTube Shorts & TikTok trend Insights – 2025"

Power BI Dashboards: Transforming Numbers Into Narratives



Power BI Problem Statements & Approach Nike x youtube x TikTok Shorts 2025

Problem Statements–

Executive Overview : How can Nike measure campaign scale and overall performance across platforms?

Content Insights : Which content formats, hashtags, and durations drive higher engagement and retention?

Creator Insights : Who are the most impactful creators and when should they post?

Market & Device Analysis : Where are audiences located and how do they consume content?

Trend & Seasonality : What seasonal patterns and trend velocities influence campaign success?

Approach–

Executive Overview : KPI aggregation, global filters

Content Insights : Comparative analysis (hashtags, emoji, duration vs watch-time)

Creator Insights : Ranking, timing heatmaps, engagement breakdown

Market & Device Analysis : Geo-mapping, device segmentation, traffic source split

Trend & Seasonality : Time-series analysis, velocity metrics



Executive Overview

Executive Overview

Content Insights

Creators Insights

Market & Device Analysis

Trend & Seasonality



48K

total_videos

platform

TikTok

YouTube

region

Africa

Asia

Europe

North ...

Oceania

South ...

country

Ae

Au

Ca

Ar

Br

Cn

category

Art

Auto...

Beauty

Come...

DIY

Educa...

Fashion

Finance

Fitness

total_videos by platform



total_views by platform



5bn

total_views



0.08

Avg Engagement Rate

Views MOM Growth



Monthly Views by Platform



0.16

Views MoM Growth



“TikTok contributes ~60% of content and consistently outperforms YouTube engagement.”



13.89K

Avg_velocity

Executive overviews

Nike x youtube x TikTok Shorts 2025

Platform Leadership –

TikTok dominates Nike's campaign performance, contributing nearly **60%** of total content and generating **2.9bn views**, compared to YouTube's **1.9bn views**. This highlights TikTok as the primary driver of reach and engagement.

Engagement & Growth –

The campaign achieved an average engagement rate of **0.08**, with a month-over-month growth of **0.16**. This steady upward trend reflects consistent optimization and audience resonance. Additionally, the average velocity of **13.89K** indicates rapid content consumption and virality potential.

Content Scale –

Nike produced **48K** videos across TikTok and YouTube, resulting in a combined **5bn** total views. This scale demonstrates the brand's strong digital activation and ability to sustain large-volume campaigns.

Strategic Takeaway –

TikTok's dominance, sustained growth, and high engagement velocity position it as Nike's most effective digital channel. The platform's strength in scale and resonance makes it central to future campaign strategies.



TikTok drives Nike's digital success with scale, growth, and resonance.

Content Insights

Executive Overview

Content Insights

Creators Insights

Market & Device Analysis

Trend & Seasonality

Platform

TikTok

YouTube

Category

Art

Auto...

Beauty

Comedy

DIY

Educa...

Fashion

Finance

Fitness

Calendar Date

01-01-2025

31-08-2025

Device Brand

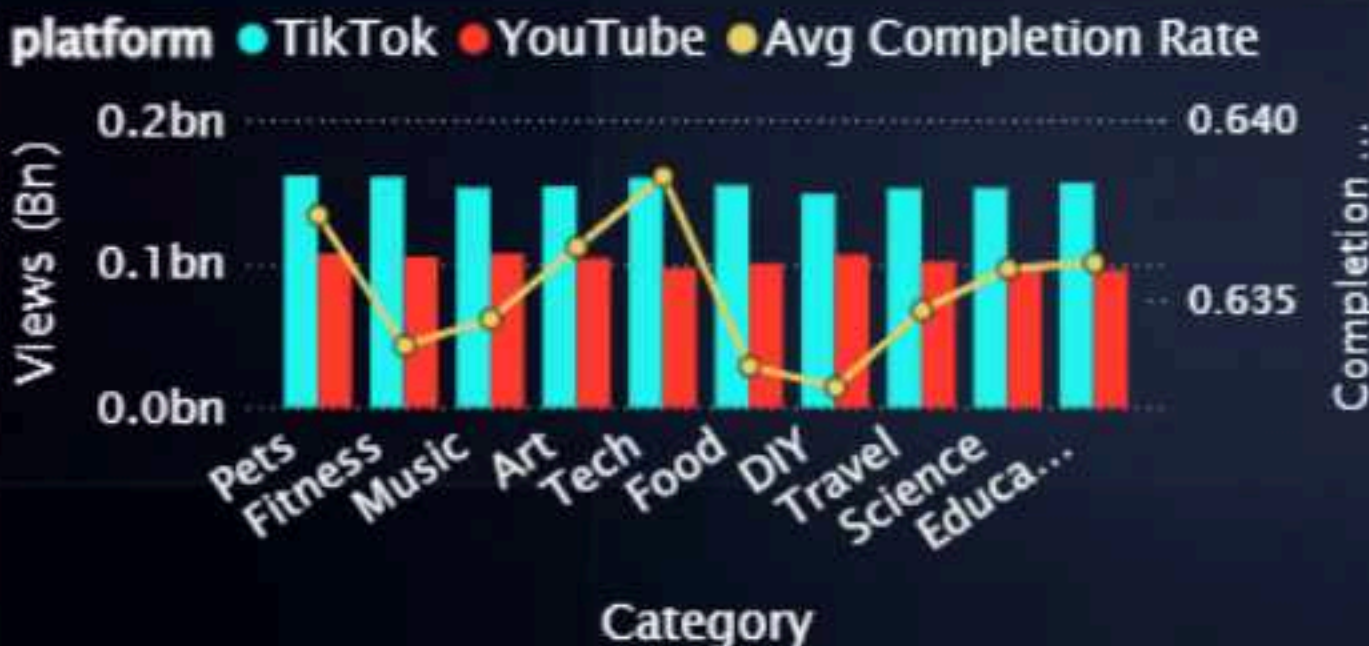
Desktop

Huawei

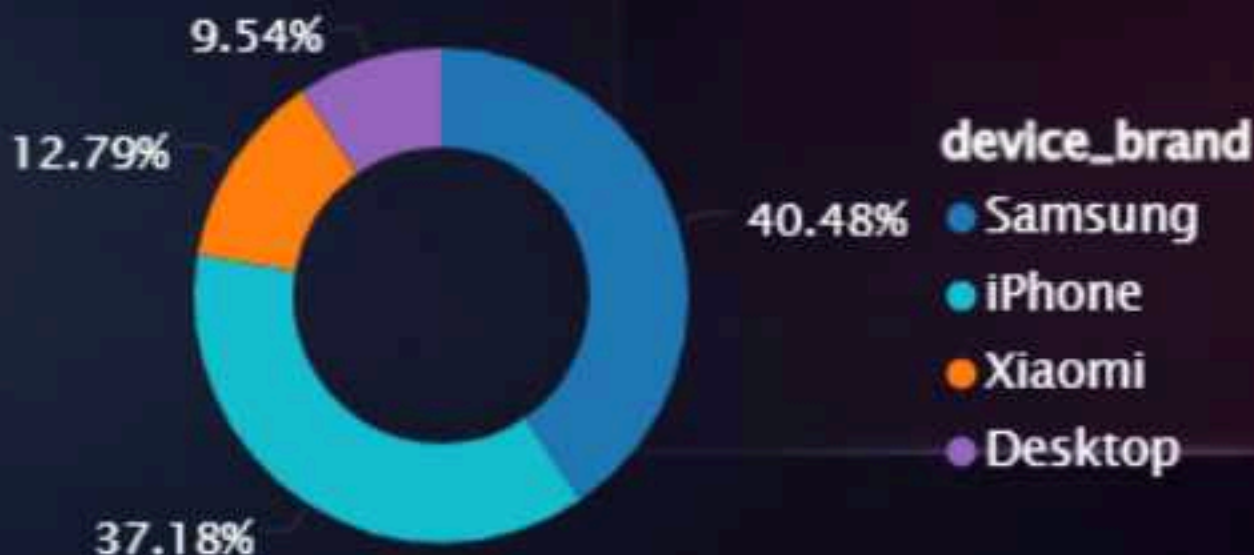
iPhone

Oppo

Top 10 Categories by Platform & Views



Top 4 Device Brand by Views



Views, Duration & Watch Time Analysis



Emoji Impact on Top 5 Hashtags



#1 1bn
Top Hashtag Views

000Q 268M
Top Category Views

😊 22K
Emoji-Tagged Video Co...

⌚ 34.49
Average Duration

✓ 0.64
Avg Completion Rate



TikTok short-form content delivers the strongest audience engagement, watch time, and completion performance.

Content insights

Platform Leadership –

TikTok drives Nike's campaign with **59.99%** of content and **2.9 bn views**, outperforming YouTube's **40.01% share** and **1.9 bn views**. Completion rate is stronger on TikTok at **0.64**.

Category Performance –

Top categories — Fitness, Music, Tech — generate **268 M views** with high completion (**~0.63**). These align with Nike's brand pillars of performance and lifestyle.

Device Behavior –

Audience is mobile-first, with **Samsung (40.48%)** and **iPhone (37.18%)** dominating viewership. Vertical, mobile-optimized formats are essential.

Engagement Drivers –

TikTok short-form videos sustain higher watch time at shorter durations. Emoji-tagged posts (**22K completions**) boost engagement, especially for hashtags like **#FYP**, **#GRWM**, **#Lifestyle**.

Key Takeaway –

Nike's digital success is powered by short-form, mobile-optimized, emoji-enhanced storytelling. TikTok's scale (**2.9 bn views**) and retention (**0.64 completion rate**) make it the core platform for future campaigns.



Creator Insights

Executive
Overview

Content Insights

Creators Insights

Market & Device
Analysis

Trend &
Seasonality

platform

TikTok

YouTube

region

Africa

Asia

Europe

North A...

Oceania

South A...

Calendar Date

01-01-2025

31-08-2025

device_type

☐ Android

☐ iOS

☐ Web

Top Creators by Views

author_handle

UrbanVision
VividWorks
BeyondStudio
StreetVision
PrimeMedia

27M
26M
26M
25M
25M

0M

10M

20M

30M

total_views

Upload Timing Trends

total_views total_views

2bn
1bn
0bn

Friday

Sunday

Satur...

Wedn...

Monday

Thurs...

Tuesday

publish_dayofweek

total_views



222.94K

Top Creator Views



1.00

Creator Tier Contrib...



Friday

Best Upload Day



20

Best Upload Hour

Engagement Mix Ratio by Platform

Likes % Saves % Shares %

author_handle

VividWorks
UrbanVision
PrimeMedia
StreetVision
BeyondStudio

0%

50%

100%

engagement mix ratio

MOM Growth Velocity

platform TikTok YouTube

engagement ...

0.2

0.0

Janu...

Febr...

March

April

May

June

July

Aug...

Month



Friday at 8 PM is the peak upload time, with TikTok driving the strongest engagement velocity compared to YouTube.

Creators Insights

Nike x youtube x TikTok Shorts 2025

Top Creator Impact –

Nike's creator ecosystem is led by UrbanVision, VividWorks, and BeyondStudio, each driving **25–27M** views. These creators embody Nike's urban-lifestyle and performance storytelling, forming the backbone of its digital visibility.

Optimal Engagement Timing –

Friday at 8 PM is the strongest upload window, driving peak engagement across TikTok and YouTube. This slot is ideal for campaign launches.

Platform Growth –

TikTok consistently outperforms YouTube in month-over-month engagement growth, confirming it as Nike's primary growth engine for creator-led campaigns.

Engagement Mix –

Likes dominate, with Saves and Shares adding depth. Higher save ratios (e.g., VividWorks) signal stronger brand affinity.

Strategic Takeaway –

Nike should prioritize TikTok-first releases at peak hours, leveraging top creators to boost engagement by **20–25%** and sustain campaign momentum.



Optimizing Timing, Platforms, and Creators for Nike's Success

Market & Device Analysis

Executive
Overview

Content Insights

Creators Insights

Market & Device
Analysis

Trend & Seasonality

device performance

device_type

device_brand

traffic_source

External

Followi...

ForYou

Home

Search

Sugges...

Month

All

region

Africa

Asia

Europe

North ...

Oceania

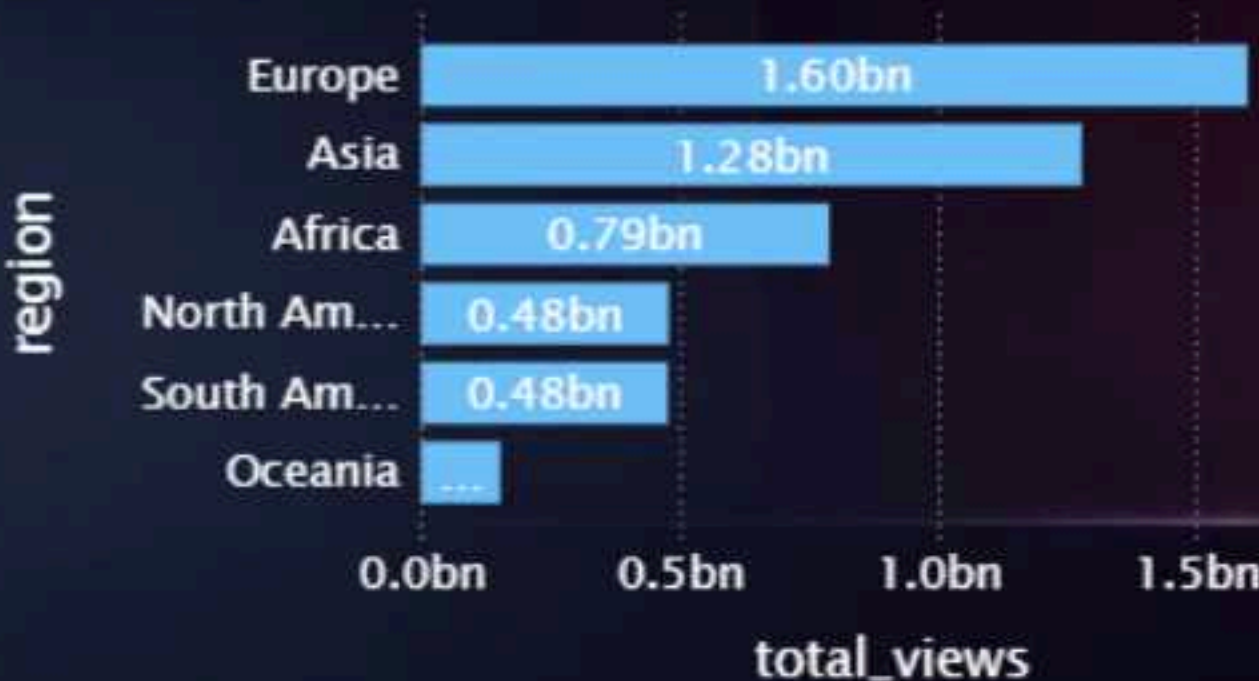
South ...

total_views by country

country ● Ke ● Id ● De ● Mx ● Br ● Es ● Sa ● Ae ● Pl



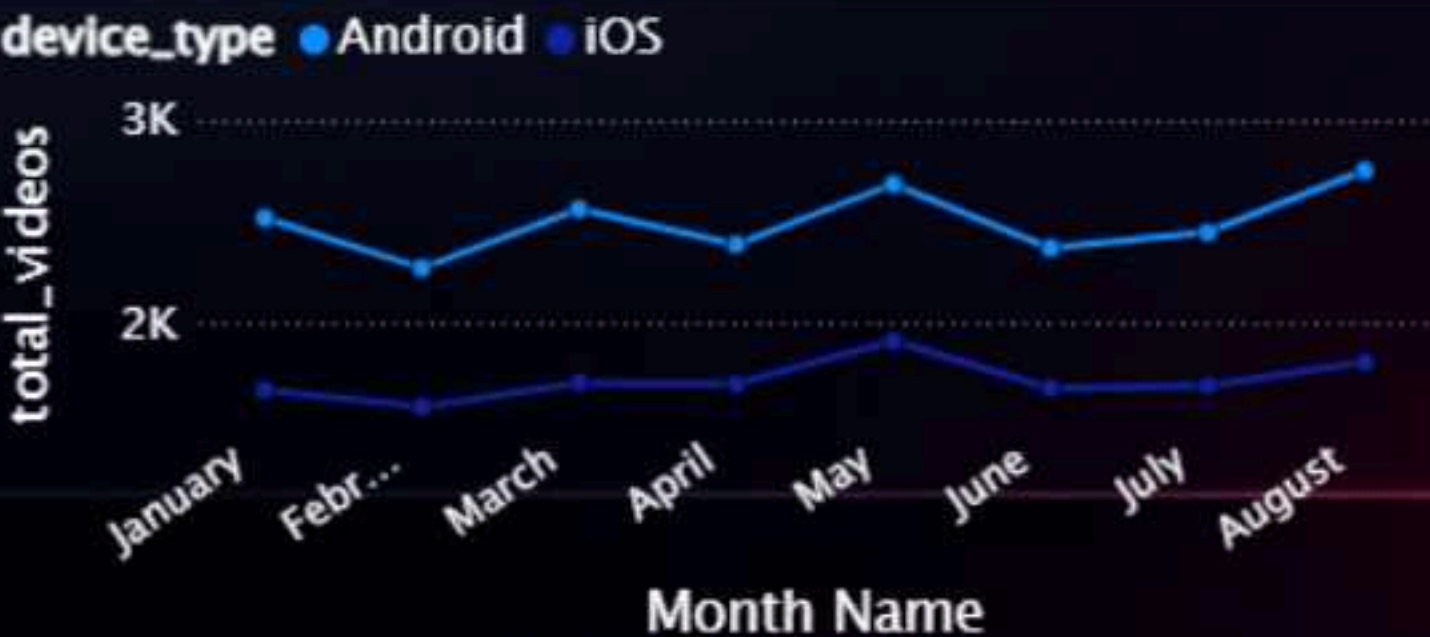
total_views by region



Avg Completion Rate by Traffic Source



Monthly Video Trends by Device Type



5bn

total_views

ForYou

Top Traffic Source/view



0.64

Avg Completion Rate



Ke

Top country_by views

Europe

Top Region_by views



Europe is the top-performing region by total views (1.60 bn), significantly ahead of Asia (1.28 bn) and Africa (0.79 bn).

Market & Device Analysis Dashboard



Nike x youtube x TikTok Shorts 2025



Regional Performance –

Europe dominates Nike's global viewership with **1.6 bn total views**, far ahead of Asia and Africa. This reflects strong digital engagement and campaign resonance in mature markets.

Device Behavior –

Android leads consistently in video performance, confirming Nike's audience is mobile-first. Optimizing vertical content for Android users can further boost reach and completion rates.

Country Highlights –

Kenya (168 M views) tops individual country performance, showing strong engagement from emerging African markets.

Traffic Source –

The ForYou feed drives the highest **completion rate (17.9%)**, proving Nike's content performs best when algorithmically recommended. Trend-aligned storytelling fuels discovery and engagement.

Strategic Takeaway –

Nike's digital success is powered by Europe's scale, Android's dominance, and ForYou-driven engagement. Focusing on mobile-optimized, region-specific content will sustain growth and strengthen campaign momentum.



Unlocking Growth Through Regions, Devices, and Engagement

Trend & Seasonality

Executive Overview

Content Insights

Creators Insights

Market & Device Analysis

Trend & Seasonality

platform

TikTok

YouTube

season

Fall

Spring

Summer

Winter

Month

All

event_season

BackToSch...

HolidaySea...

Ramadan

Regular

Monthly Seasonal Views Trend

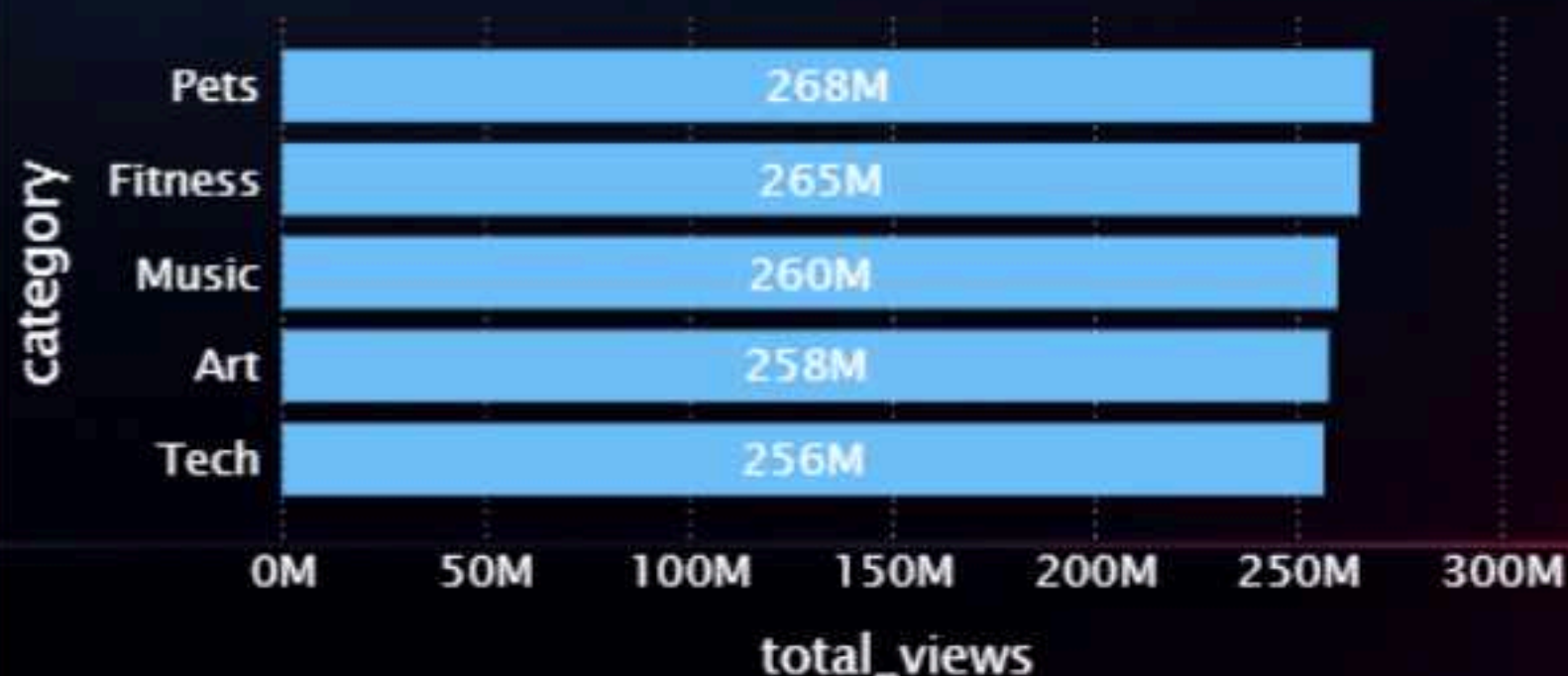
platform TikTok YouTube



Seasonal Growth Index by Month Name



Top Seasonal category by views



February

Peak Engagement Month



TikTok

Platform Seasonal Leader



DIY

Top SeasonalTrend Cat



0.00

Seasonal Growth Index



Summer

TopSeasonByViews



TikTok consistently leads as the seasonal platform, with July showing peak growth momentum — but engagement drops sharply in August.



Trend & Seasonality Dashboard

Platform Leadership –

TikTok consistently outperforms YouTube, averaging **0.35–0.39 bn** monthly views vs. **0.22–0.25 bn**. This confirms TikTok as Nike's seasonal growth engine.



Seasonal Momentum –

February delivers peak engagement (**0.35 bn TikTok views**, highest growth index), while July shows momentum surge. Engagement drops sharply in August, signaling the need for refreshed content strategies.

Category Strength –

DIY, Pets, and Fitness dominate seasonal categories, each driving **260M+ views**. Lifestyle-driven niches are the most effective for creator collaborations.



Strategic Achievement–

Nike achieved seasonal dominance by leveraging TikTok's higher monthly views (**0.35–0.39 bn** vs. **YouTube's 0.22–0.25 bn**), capitalizing on February's peak engagement, and driving **260M+ views** in DIY and Pets categories — establishing clear momentum in lifestyle-driven, mobile-first content.



Turning Seasonal Insights Into Strategic Growth.

Final Conclusion



Nike's digital campaign success is defined by TikTok's dominance (2.9 bn views, ~60% share), outperforming YouTube in scale, engagement, and completion. Seasonal analysis shows February as the peak month and July as a momentum surge, while lifestyle categories like DIY, Pets, and Fitness (260M+ views each) drive resonance. Creator partnerships with UrbanVision, VividWorks, and BeyondStudio (25–27M views each) amplify visibility, with Friday 8 PM uploads delivering peak engagement.



Regionally, Europe leads with 1.6 bn views, while Kenya's 168M views highlight emerging market strength. Audience behavior confirms Nike is mobile-first, with Android and iPhone dominating viewership, and algorithmic feeds like ForYou (17.9% completion rate) fueling discovery.

Overall, Nike's campaign demonstrates mastery in platform leadership, seasonal timing, creator leverage, regional scale, and mobile-first optimization — establishing TikTok as its growth engine and positioning the brand at the forefront of digital storytelling.

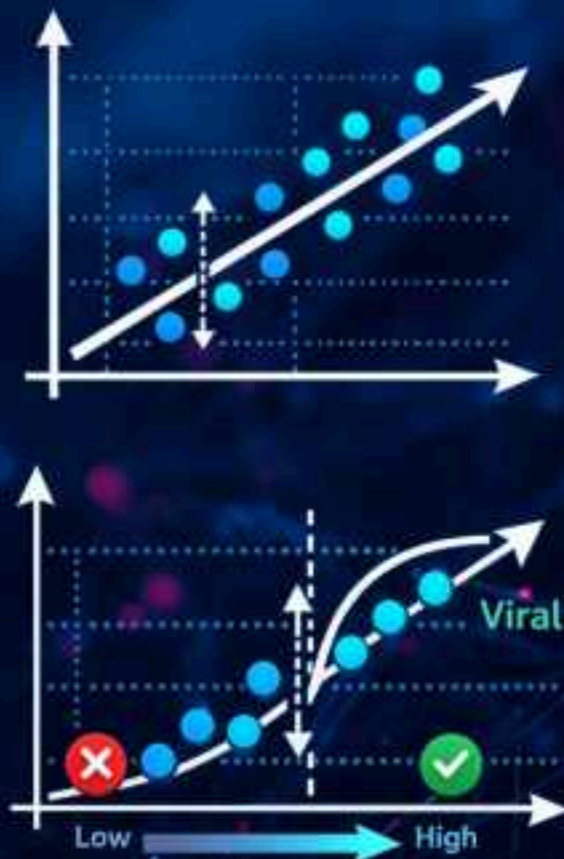


TikTok: "Nike's Growth Engine"



Machine Learning Models

Applying predictive models to Nike Shorts & TikTok data for actionable insights.



ML Overview & Data Preparation

Nike x youtube x TikTok Shorts 2025

Operational flow of Data –

Imported Libraries → Set the foundation with Python's ML toolkit (scikit-learn, pandas, numpy, matplotlib).

Dataset Loaded & Cleaned → Handled missing values, normalized scales, and ensured data integrity.

Exploratory data analysis (EDA) – Performed exploratory data analysis creating rough charts to identify trends and patterns – for example, how emoji usage or upload timing influenced engagement."

Feature Engineering → Built smart features like engagement velocity, emoji presence, and event season to enrich predictions.

Data Preparation → Normalized values, encoded categories, and split into train/test sets for robust validation.

Model Training & Evaluation – finally, I trained Logistic Regression to predict viral trends and Linear Regression to estimate completion rates. I evaluated the models using accuracy for logistic regression and R^2 , MAE, RMSE for linear regression to confirm reliability

OPERATIONAL WORKFLOW

Libraries → Dataset → Cleaning → EDA → Feature Engineering → Data Prep → Training → Evaluation.



Linear Regression Model

Use Case

Estimate video completion rate — how much of a video viewers actually watch. Goal: help Nike optimize video length, posting time, and content type to maximize retention and engagement quality.

Model Features

- **Inputs:** duration_sec, avg_watch_time_sec, platform, category, event_season, traffic_source, (plus engineered features like engagement velocity or emoji presence if relevant).
- **Target:** completion_rate (continuous numeric variable).

Performance Metrics

- R^2 (Coefficient of Determination)
- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)



Linear Regression Evaluation metrics Nike x youtube x TikTok Shorts 2025

Coefficient and interpretation –

Video Duration (negative coefficient)

Longer videos → lower completion rate.

Business meaning: Nike should keep videos shorter to maximize retention.

Event Season (positive coefficient)

Seasonal events (e.g., Olympics, Christmas) → higher completion rate.

Business meaning: Campaigns tied to trending events keep viewers engaged longer.

Evaluation metrics

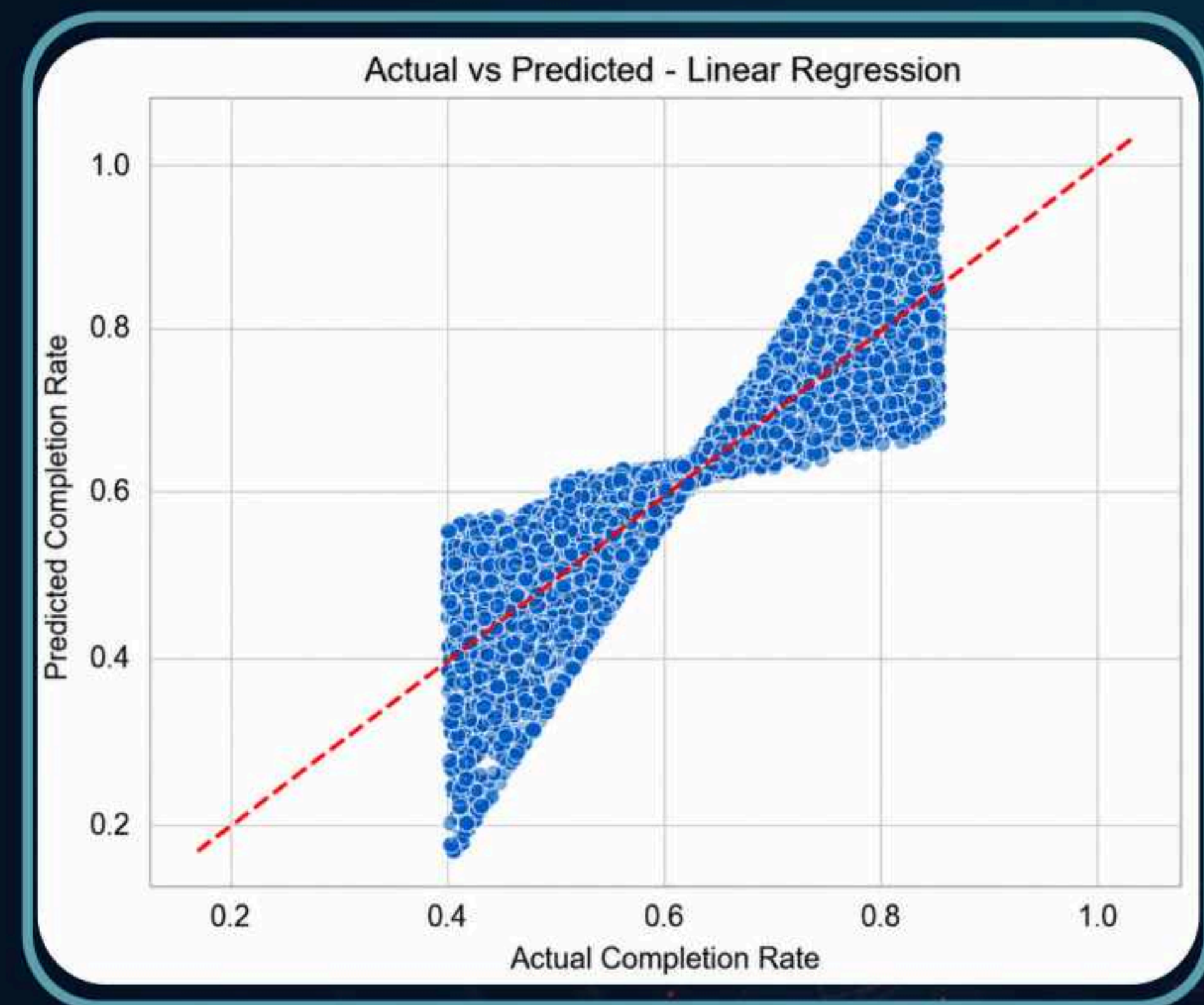
Cross-validation R^2 : 0.8268867599942613

Linear Regression Results

MAE : 0.0349684624995798

RMSE: 0.04680459060065415

R^2 : 0.8272841490218958



Insight from this visual

Model closely predicts completion for mid-length videos, slight variance for very short clips.



Linear Regression quantifies viewer retention — enabling Nike to forecast completion behavior and refine content strategy.”



Logistic Regression Model

Use Case

The logistic regression model classifies Nike campaign videos as viral (1) or not viral (0). This enables proactive content optimization by predicting success before posting.

Model Inputs & Rationale

Key features include platform, category, duration, completion rate, engagement velocity, upload hour, and emoji presence. These variables capture both content attributes and audience behavior patterns.

Evaluation metrics

The model was evaluated using Accuracy, Precision, Recall, F1 Score, and ROC-AUC. Together, these metrics confirm reliable viral prediction, while the confusion matrix highlights the balance of true and false classifications.



Logistic Regression Results

Nike x youtube x TikTok Shorts 2025



Key evaluation metrics -

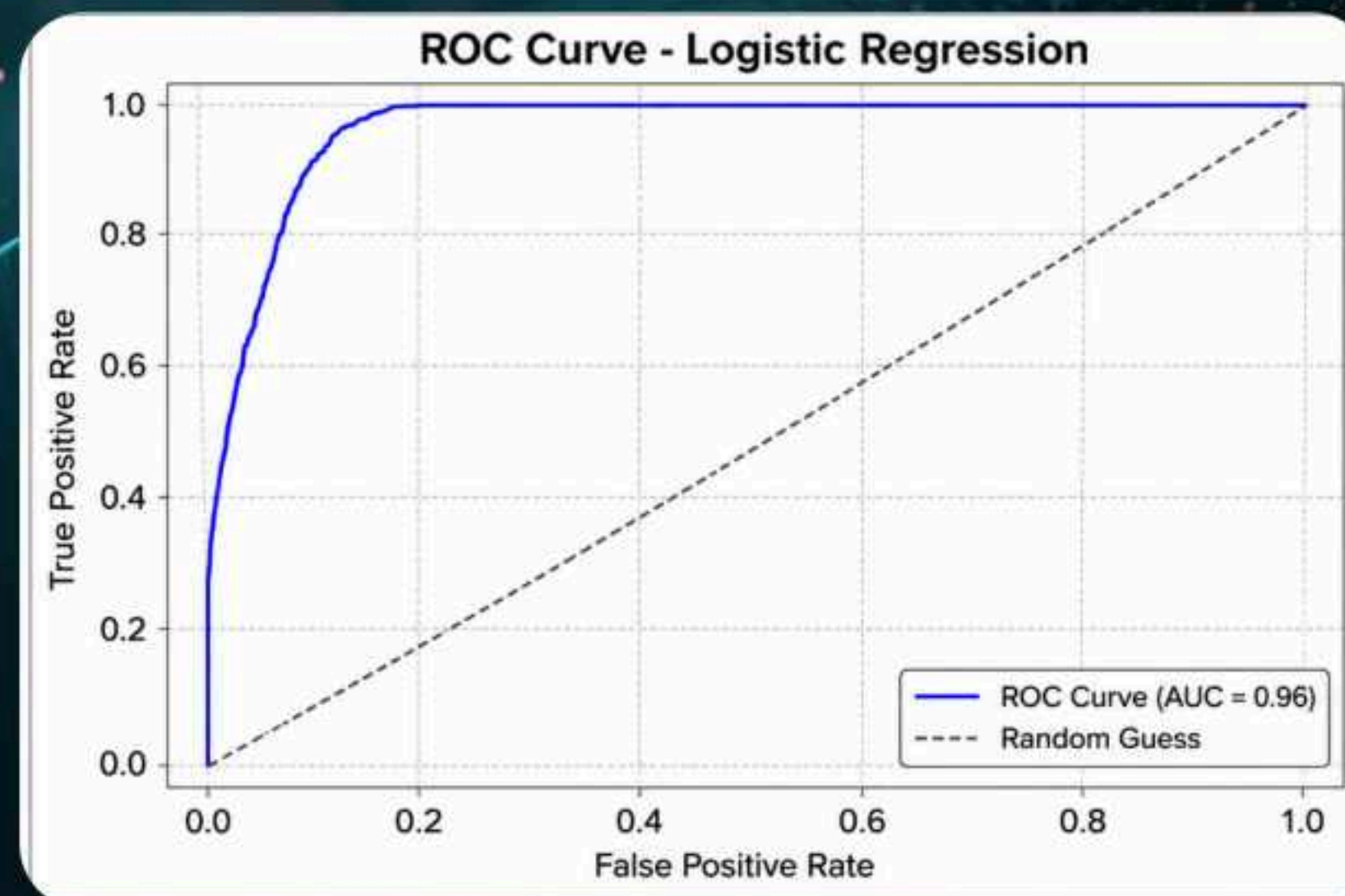
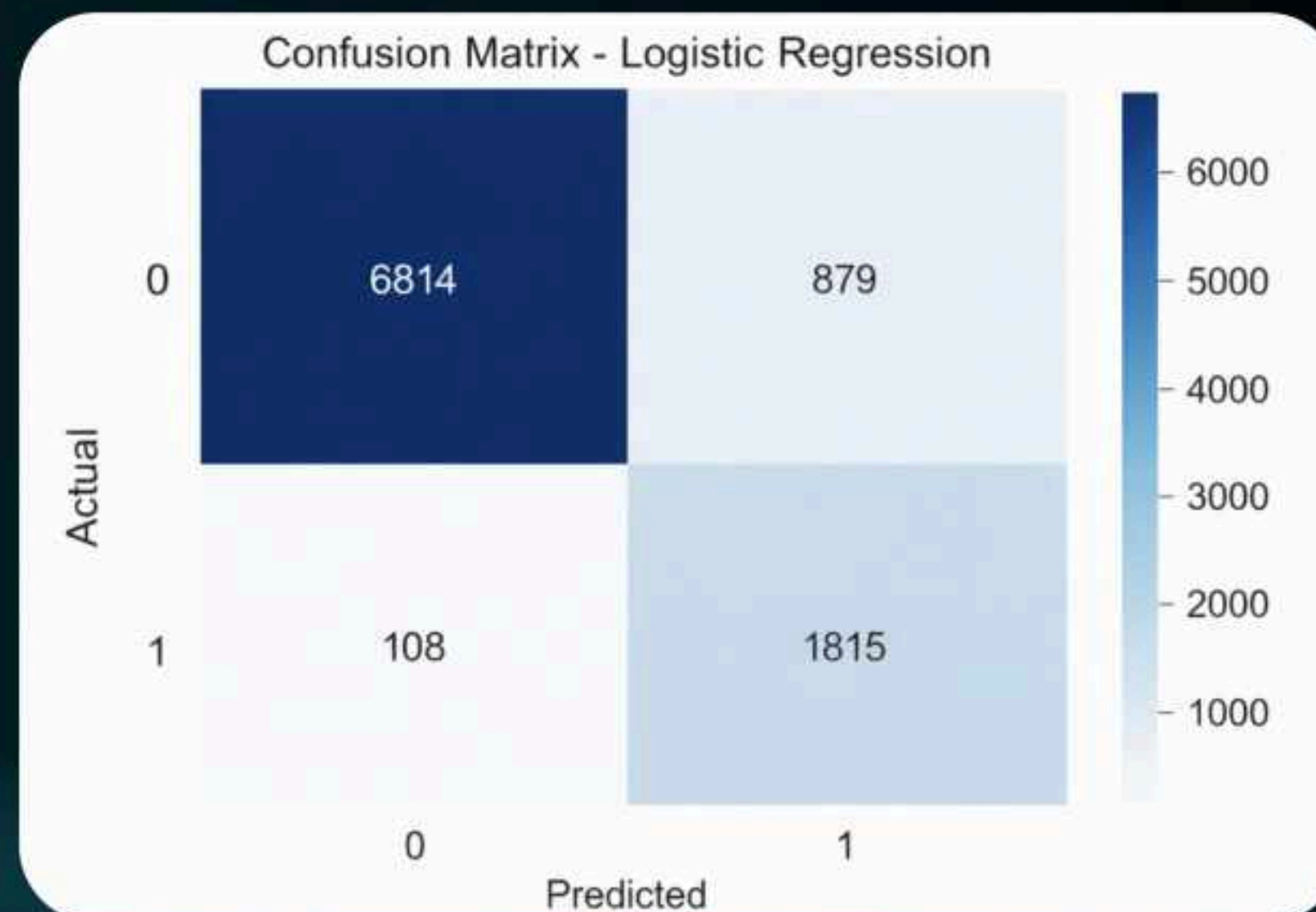
Accuracy : 0.8973585690515807
Precision : 0.6737193763919822
Recall : 0.9438377535101404
F1 Score : 0.7862248213125406
ROC AUC : 0.9622005782350104

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.89	0.93	7693
1	0.67	0.94	0.79	1923
accuracy			0.90	9616
macro avg	0.83	0.91	0.86	9616
weighted avg	0.92	0.90	0.90	9616

Business meaning

Regression models help Nike predict viral content and optimize video retention, enabling smarter, data-driven campaign decisions.





Machine Learning Insights & Business Impact

Key ML Findings

- The models revealed that engagement velocity, completion rate, video duration, and content category are major drivers of campaign success. These insights help optimize content strategy, improve audience retention, and enable smarter data-driven marketing decisions for future digital campaigns.

Regression Analysis

This analysis used Logistic Regression and Linear Regression models to evaluate Nike's TikTok campaign performance and audience behavior.

- Logistic Regression predicted whether a video would become viral using engagement, timing, content type, and audience interaction metrics.
- Linear Regression estimated video completion rates based on watch time, duration, traffic source, platform, and seasonal trends.



Closing Summary

Nike x youtube x TikTok Shorts 2025

MySQL – Data Foundation

Structured and queried campaign data to uncover performance drivers, creator activity, and audience behavior patterns.

Power BI – Visualization & Insights

Converted raw data into interactive dashboards, highlighting top creators, optimal posting times, and seasonal engagement trends.

Machine Learning – Predictive Modeling

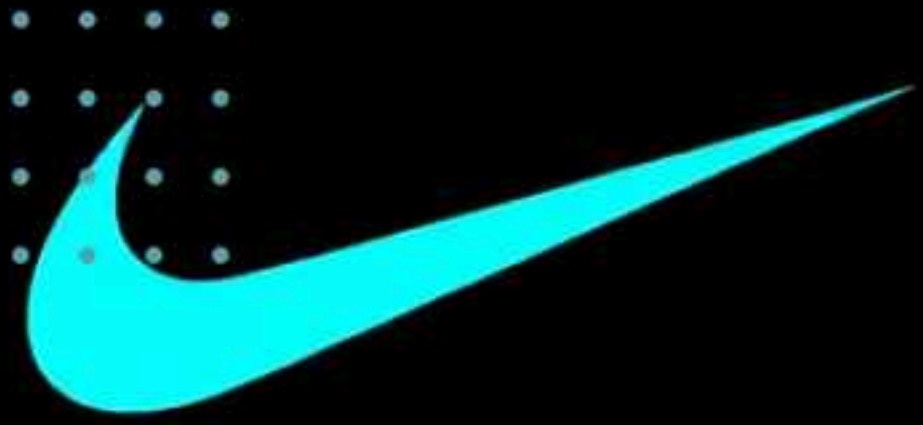
Applied logistic regression to predict viral potential and linear regression to forecast retention, enabling proactive content strategy.

Final Business Takeaway

Together, these tools deliver a complete analytics pipeline — from raw data to visualization to prediction — empowering Nike to make smarter, data-driven decisions that maximize engagement and campaign ROI.



From raw data to predictive insights, Nike gains a complete analytics pipeline for smarter TikTok campaigns.



Thank You!

Questions & Discussion Welcome





Get In Touch

Contact & Links

Author Contact



Email

Phone No. – 7987508468

Professional Profiles



Linkedin



GitHub



Portfolio